ClaimsGuru™ Enterprise AI Platform

BACKEND TECHNOLOGY STACK SPECIFICATION

Target Infrastructure: Microsoft Azure Cloud	**Core Framework: Python 3.11 + FastAPI**	**Primary LLM: Azure OpenAI GPT-4o**
Architecture: Async Celery Microservices	**Enterprise DB: Azure SQL Database (MSSQL)**	**Security & IAM: Microsoft Entra External ID**

1. Executive Summary: What Powers ClaimsGuru?

ClaimsGuru is an intelligent health insurance claim auditing and adjudication platform built to automate reimbursement claims processing. Running natively on **Microsoft Azure**, it converts messy, scanned, multi-page hospital bills, diagnostic receipts, and discharge summaries into structured, verified, and audited expense tables.

The backend is built with an **asynchronous microservices architecture** powered by **FastAPI**, **Celery**, and **Azure Cache for Redis**. Claim extraction leverages direct parallel calls to **Azure OpenAI GPT-4o** alongside **Azure AI Document Intelligence** for computer vision table parsing, delivering lightning-fast turnaround times with complete mathematical accuracy.

2. Enterprise Value Pillars

■ High-Throughput Async Engine Processes multi-page hospital bills simultaneously with zero UI lag using distributed Celery background workers and in-memory Redis message queues.	**■ Precision Extraction** Combines Azure Document Intelligence layout analysis with Azure OpenAI GPT-4o parallel semantic parsing to extract complex hospital tariffs and line items accurately.
■■ Bank-Grade Security & Isolation Protected with Microsoft Entra External ID (OAuth2/OIDC), private Azure Virtual Networks, and hardware-backed Azure Key Vault secret encryption.	**■ Elastic Cloud Scalability** Containerized microservices running on Azure Container Apps with automated autoscaling based on incoming claim traffic volumes.

3. Verified Backend Technology Stack Specifications

The following table details the technologies actively used in the ClaimsGuru backend:

Architectural Tier	Core Technology	Exact Version	Role & Implementation in Platform
API Gateway & Services	Python & FastAPI	**Python 3.11+ FastAPI 0.128**	Asynchronous, low-latency REST endpoints handling claim lifecycle, document uploads, and real-time status polling.
ASGI Server Engine	Uvicorn + uvloop	**uvicorn 0.39 uvloop 0.21**	High-performance event-loop server built for enterprise multi-tenancy and high concurrency.
Task Queue & Pipeline	Celery Distributed Queue	**celery 5.4.0**	Orchestrates deep AI audit pipelines asynchronously so users never experience UI blocking or timeouts.
In-Memory Cache & Broker	Azure Cache for Redis	**redis 5.2.1**	Millisecond state synchronization for real-time progress bars, claim session caching, and task queuing.
Enterprise Event Messaging	Azure Service Bus	**SDK 7.14.0**	Fault-tolerant, guaranteed message delivery and event processing across claim ingress and worker nodes.
Enterprise Database	Azure SQL Database (MSSQL)	**SQLAlchemy 2.0 pyodbc / pymssql**	ACID-compliant storage for claims, hospital data, ICD-10 medical records, and audit findings with connection pooling.

Schema Migrations	Alembic	**alembic 1.18**	Automated, version-controlled relational database schema updates and tracking.
Secure Document Lake	Azure Blob Storage	**StorageV2 (Hot) TLS 1.2+ HTTPS**	Encrypted object storage holding raw medical records, scanned receipts, and generated audit report PDFs.
Generative AI & LLM	Azure OpenAI Service	**GPT-4o (API: 2024-11-20)**	Direct asynchronous REST integration for semantic table parsing, clinical entity extraction, and rate-limit backoff.
Computer Vision & OCR	Azure AI Document Intelligence	**API 2023-07-31 Form Recognizer**	Deep layout parsing, table structure extraction, and bounding-box coordinates for Indian hospital bills.
PDF Parsing & Rasterizing	PyMuPDF & pdfplumber	**pymupdf 1.25 pdfplumber 0.11**	High-speed digital text extraction, DPI-optimized rendering, and vector line detection for medical PDFs.
Semantic Catalog Search	FAISS Vector DB + BM25	**faiss-cpu 1.9 rank-bm25 0.2**	Hybrid lexical and semantic lookup of standard ICD-10 medical diagnostic codes and procedure master packages.
Predictive Analytics	XGBoost & LightGBM	**xgboost 2.0+ lightgbm 4.0+**	Machine-learning risk assessment flagging duplicate claims and billing anomalies.
Enterprise Identity (IAM)	Microsoft Entra External ID	**CIAM OAuth2 OIDC & RS256 JWT**	Single Sign-On (SSO) and Role-Based Access Control (RBAC) ensuring only authorized claim officers access sensitive data.
Secrets & Privacy Vault	Azure Key Vault	**Hardware HSM**	Centralized cryptographic key and secret management meeting strict enterprise compliance standards.
Email & Notifications	Azure Communication Services	**ACS Email SDK**	Automated transactional email dispatch notifying claimants and claim officers the moment an audit is ready.
Automated Report Engine	WeasyPrint + Jinja2	**weasyprint 63+ Jinja2 3.1**	Generates official, print-ready, pixel-perfect PDF Audit Reports with breakdown charts and regulatory citations.

Verification Note: *All technologies, libraries, and cloud services in this document reflect the active codebase and live deployment in the claimsGuru Microsoft Azure pre-production environment.*